



Sri Lanka Institute of Information Technology

Report on Industrial Training

At

Divor Automations (Pvt) Ltd Company

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Abstract

To complete the four-year degree requirements, all undergraduate students in the Electrical and Electronics Engineering department must undergo a 12-week industrial training program. From June 16 to September 5, 2025, I had the privilege of carrying out my industrial training at **Divor Automations (Pvt) Ltd**, an innovative IoT-based project development company in Sri Lanka. During this training, I was exposed to a wide range of practical applications, including IoT device manufacturing, PCB designing and fabrication using laser cutting for single-layer boards, microcontroller programming, panel board assembly, and automation solutions. I also gained valuable knowledge in integrating embedded systems with IoT platforms, customizing industrial control systems, and learning about SCADA/HMI applications. This training not only enhanced my technical skills but also provided me with real-world exposure to the challenges and solutions in industrial automation and IoT integration. The knowledge and experience I acquired during this period are summarized in this report.

Acknowledgement

Using this wonderful opportunity, I would like to sincerely thank everyone who supported and guided me during my industrial training program at **Divor Automations (Pvt) Ltd.**

First, I would like to express my gratitude to the **SLIIT Training Division** for arranging this valuable opportunity for students to gain practical industry experience alongside academic learning.

I am deeply thankful to the **Chief Executive Officer of Divor Automations, Mr. Anjana Vishvanath**, who served as my training supervisor. His constant guidance, encouragement, and willingness to share his expertise in IoT system design, PCB fabrication, embedded programming, and automation solutions made my training experience both enriching and inspiring.

I also extend my appreciation to the **Directors, Technical Managers, Engineers, and Project Supervisors** at Divor Automations for their invaluable assistance and knowledge sharing throughout my training. Their mentorship not only enhanced my technical knowledge but also gave me insights into real-world industrial challenges and solutions.

Additionally, I wish to thank all the staff and team members at Divor Automations for their cooperation and support, which contributed immensely to the success of my training.

Lastly, I would like to thank everyone mentioned above, as well as all others who supported me in any way, for making this training experience a memorable and fruitful part of my academic journey.

List of Abbreviations

MCU	: Micro Controller Unit
PCB	: Printer Circuit Board
SMT	: Surface Mount Technology
THT	: Through Hole Technology
OTA	: Over The Air
PID	: Proportional Integral Derivative
TDS	: Total Dissolved Solids
PWM	: Pulse Width Modulation
ADC	: Analog-to-Digital Converter
QFN	: Quad Flat No-Lead
GPIO	: General Purpose Input/Output
FDTI	: Future Technology Devices International
MPU	: Motion Processing Unit

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1.0 INTRODUCTION TO TRAINING ESTABLISHMENT

1.1 Introduction



Figure 1.1 Company Logo

Telephone : [+94 77 7332307](tel:+94777332307)

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Divor Automations Private Limited is a recently established company formed on the basis of bringing advanced technology to Sri Lanka's industry. Its goal is to revolutionize the domestic information technology sector by offering products that are reasonably priced and made according to industry standards and specifications, as requested by clients.

1.2 Vision and Mission

Vision: *“Your Partner in Digital Revolution”* – focusing on revolutionizing the local tech industry through innovation and advanced solutions.

Mission: To provide simple, cost-effective, and customer-oriented products and services across industries while empowering clients through digital transformation and automation

1.3 Company Background/History

Sectors served: Textile, Avionics, Biomedical, Pharmaceutical, Hotel, and Construction industries.

Expertise: PCB designing, panel boards, control systems, HMI & SCADA systems, IoT device manufacturing, software integration, and mechanical systems.

Growth Path: Established as a tech-driven startup in Sri Lanka, Divor Automations has expanded its portfolio to include industrial automation equipment, IoT solutions, training simulators, home automation, and BMS systems.

Recognition: The company has won multiple awards such as E-Swabhimani 2020 (Winner – Inclusion & Empowerment), SLT Zero One Awards 2020, and National ICT Awards (NBQSA) 2020.

2.0 TRAINING EXPERIENCE IN MANAGEMENT

2.1 Organizational management structures and reporting procedures.

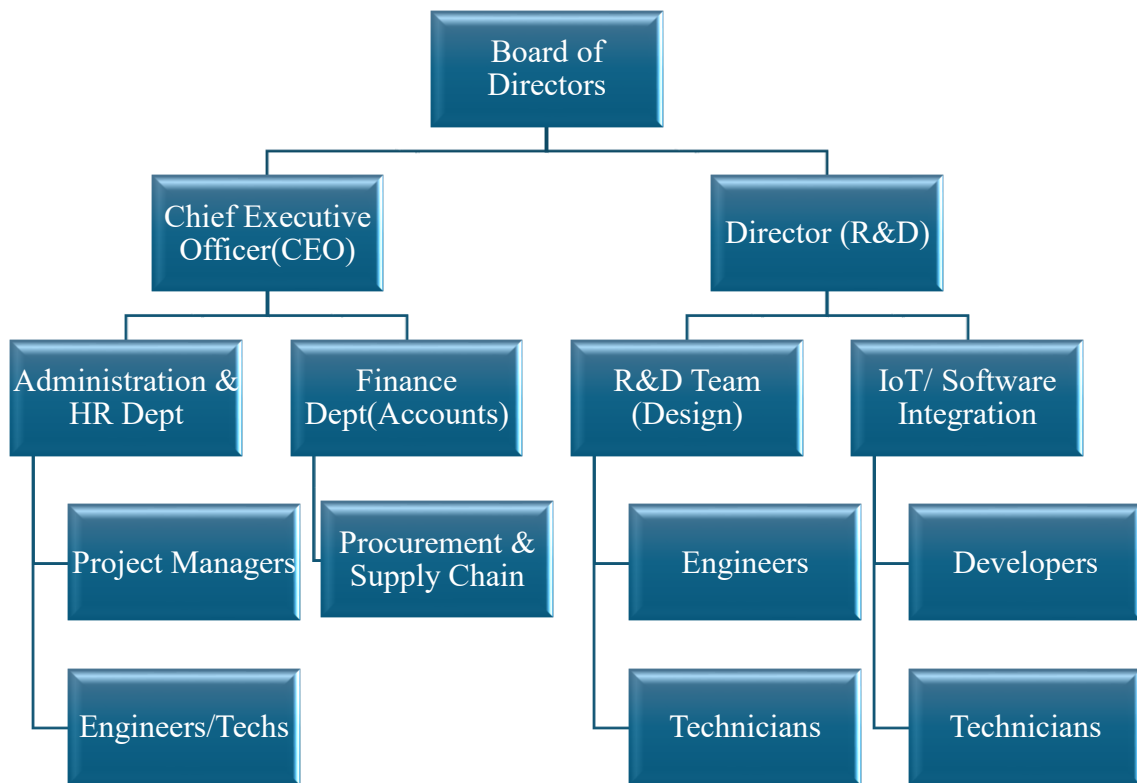


Chart 1.1 Management Structure

2.2 Work progress monitoring techniques.

The organization tracks work progress through an output-oriented method.

Major practices are:

Engineers and technicians use online tools (typically project management software like Trello or Asana) to allocate and track daily tasks.

Weekly Review Meetings: Each department reports on works in progress, mentioning milestones reached and work still pending.

Performance Indicators (KPIs): Productivity is monitored through measures like system performance, purchasing efficiency, and project timing.

Client Feedback Loops: Deliverables at each level are double-checked against client expectations in order to ensure alignment.

In total, project efficiency improves, accountability is increased, and lateness is reduced because of this systematic monitoring.

2.3 Administrative practices

Administrative procedures of Divor Automations are set to reconcile systematic administration with innovativeness. The following key practices were followed during training:

Documentation and Record Maintenance: All project reports, technical designs, and procurement information are kept electronically for future use.

Resource Allocation: Administrative personnel make sure that the necessary instruments, electronic components, and test equipment are provided on schedule to all departments.

Compliance & Quality Assurance: Strict compliance with safety regulations, industrial standards, and quality assurance procedures.

Communication Management: Enhanced communication internally through email, instant messaging tools, and periodic briefing to create an open channel of communication.

Employee Support: HR ensures the employees and engineers receive proper onboarding, training, and welfare facilities.

3.0 TRAINING EXPERIENCE IN TECHNICAL PROFICIENCY

3.1 Involvement in projects

During my industrial training at Divor Automations Pvt. Ltd., I had the opportunity to engage in several ongoing projects, which gave me an excellent insight into practical engineering methods. The projects covered various aspects, such as PCB design, industrial automation, IoT-enabled agricultural systems, and smart monitoring systems. I gained practical technical experience in all projects, since they each involved elements, including some combination of software programming, hardware design, and system integration.

3.1.1 Hydroponic Grow Box

One of the most interesting projects I experienced was the Hydroponic Grow Box, an automated plant growth, smart farming solution that allows for plant to grow without soil. The grow box used a variety of sensors including a temperature sensor, humidity sensor, pH meter, and water level sensor all connected to a microcontroller (ESP32). The data the sensors collected was sent to a cloud using IoT technology where users of the grow box could see in real-time the growth of the plants using dashboards.

My involvement in this project included:

- Supporting the calibration of the sensors for accurately measuring the environmental variables.
- Contributing to the PCB design that housed the control circuitry for the Grow Box.
- Helping with the implementation of the automatic lighting and irrigation system that uses algorithms in software to control the functions.
- The Hydroponic Grow Box project used IoT in practical use in the field of precision agriculture and addressed the need for sustainable practices and technology in agriculture.

3.1.2 IoT-Based Dashboards

The IoT-based production dashboards for the textile and apparel industry are another important project that I worked on. These dashboards collected real-time production data, such as equipment productivity, labor efficiency, and quality control metrics.

My contributions included:

- assisting the team with an IoT sensor-based data collection module.
- learning about data transfer using the MQTT protocol in the cloud.
- observing install user-friendly dashboard interfaces that displayed relevant production data on large-format LED screens.

This project introduced me to industrial data visualization and demonstrated how the Internet of Things can improve manufacturing operational efficiency tremendously.

3.1.3 Design and Development of PCBs

Additionally, I actively participated in the PCB design process for a number of automation devices. I assisted with component placement, routing, and schematic design using EasyEDA CAD tools. The created PCBs were utilized in embedded device prototypes, IoT monitoring systems, and hydroponics controllers, among other tasks.

Important lessons learnt:

- Designing with noise reduction and signal integrity in mind.
- Recognizing the significance of decoupling capacitors and ground planes.
- Soldering through-hole/SMD components and conducting continuity checks.

My knowledge of electronic design was strengthened by this experience, which linked theoretical circuit design with real-world application.

3.1.4 E-Kanban and DAV 1.0 are two prototypes of industrial automation.

I was able to observe and respond to industrial automation prototypes.

- The Driverless Automated Vehicle (DAV 1.0) is designed for material handling in a factory. I have seen its navigation system in action using sensors and a control algorithm and assisted with some straightforward testing.
- The E-Kanban Smart Stock System is an intelligent inventory management system that reduces paper-based processes. I learned about RFID-based stock tracking, while also assisting with hardware configuration.

Both projects confirmed the role of automation in current-day businesses and illustrated how small innovations can lead to large productivity improvements.

3.2 Theoretical engineering concepts with practical applications.

One of the most important aspects of my training was the chance to apply engineering theories developed in the classroom, to an industry technical project. Whilst both academic classes gave me theoretical knowledge about electronics, control systems, programming and the IoT, the training that I received at Divor Automations allowed me to see and apply these theories in the real world.

3.2.1 Electronics & Circuit Theory

I have previously studied power distribution concepts, Kirchhoff's Laws, Thevenin's and Norton's theorems, and Ohm's Law. These were applied in real-world scenarios during:

- PCB design for IoT devices requiring appropriate voltage control circuitry and sensor biasing.
- The use of voltage regulators and buck converters to stabilize power sources for hydroponic control systems.
- The use of decoupling capacitors in circuit design to reduce voltage swings and eliminate microcontroller resets.

As an example, I was involved in calibrating a water level sensor during the hydroponics project. I used concepts of voltage dividers and sensor linearization to achieve accurate analogue values for the ESP32 microcontroller from the sensor.

3.2.2 Systems of Control

According to the control theory, PID (Proportional–Integral–Derivative) controllers are applied to control a dynamic system. This notion was utilized directly in:

- The Hydroponics Grow Box system regulated the activation of the water pump using a PID algorithm based on measurements of soil moisture and water level.
- PID control provided a key component of consistent and fluid motion for motor control applications in the DAV 1.0 automated vehicle.

Application Example: Instead of immediately turning the cooling fan on or off, the control algorithm changed the speed gradually in proportion to the temperature measurement of the Grow Box that was above a predetermined threshold. This enhanced energy efficiency and reduced the stress of the system.

3.2.3 Protocols for IoT and Communication

From a theoretical focus, I considered IoT protocols such as MQTT and HTTP, as well as wireless communication protocols such as Wi-Fi, Bluetooth, and Zigbee. In practice, I worked with and observed these in:

- Real-time dashboard updates in textiles using MQTT-based communication.
- Controllers enabling ESP32 Wi-Fi capability in sending sensor data to a cloud server in hydroponics.
- RFID technology deployed for tracking inventory in the E-Kanban project.

Because of this, I now have a better sense of latency challenges, data packet loss, and reconnection methods, which is not normally emphasized in academic literature.

3.2.4 Embedded Systems

In practice, I was able to apply the academic knowledge acquired during my study of embedded C programming and microcontrollers:

- All relays, sensors, and actuators were controlled by ESP32 and Arduino microcontrollers.
- While testing PCBs, we utilized interrupting handling, particularly to read encoders for control of motors.
- I learned to take theoretical flowcharts and, through real-world debugging that relied on serial monitors, convert them into functioning firmware.

For example, many functions were executed by the ESP32 in the GrowBox project, including data upload via Wi-Fi, turning on the lights, and reading humidity sensors. Testing the real-time multitasking concepts, we learnt in theory was a reality.

3.2.5 Principles of Software Engineering

I was first introduced to data structures, modular design, and structured programming in an academic classroom. In practice:

- I applied these concepts while developing modular hydroponic controller code in Python and for Arduino.
- Within dashboard systems, incoming sensor data was organized into arrays and projected onto user interfaces through techniques for managing data.
- The concepts of software optimization were applied to manage memory use on microcontrollers with limited RAM.

3.2.6 Outcomes of Melding Theory and Practice

While developing applications, I had an insight:

Theory without comprehension of the constraints of the real world (noise, precision of sensors, connections) is not sufficient.

Engineering projects require iterative testing, where the theoretical design is modified and retested until a reliable result emerges.

My capacity for analysis and problem-solving was much improved by applying my academic knowledge to practical projects.

3.3 Design, development, and testing participated in observed

One of the primary learning outcomes of my course was gaining insight into the engineering design cycle, which involved the processes of ideation, design and development, and exemplary testing and validation. Projects such as the Hydroponics Grow Box, IoT-enabled dashboards, and PCBs for automations systems all that facilitated this process.

3.3.1 Design Phase

Starting the design process involved collecting requirements, drawing schematics, and brainstorming. For example, the team identified the essential water level, temperature, humidity, and light level in the Hydroponics Grow Box. The following design tasks were conducted in accordance with those requirements.

- **Blocks:** The system block diagram displays an ESP32 controller, sensors, actuators (lights, fans, pumps), and cloud integration.
- **PCB Design:** In EasyEDA, we designed circuits focusing on voltage regulation, sensor interface, and relay driving circuits.
- **Component Approval:** We selected accurate DHT22 temperature/humidity sensors known for their reliability and chose low-power ESP32 modules for IoT connectivity.

I contributed and understood how these schematics are prepared, how CAD layouts are generated from theoretical circuit diagrams, while considering ground planes, track width, ground planes, and noise minimization.

3.3.2 Phase of Development

Once the design was finished, the next stage was developing the software and hardware.

Development of Hardware:

- At the PCB fabrication and assembly stage, I utilized soldering stations.
- I was able to perform both SMD soldering with hot plates and through-hole soldering.
- I produced a single-layer PCB for the GrowBox to enable sensors and actuators to be controlled.

Development of Software:

- The ESP32 microcontroller was coded using the Arduino IDE with my assistance.
- The code included Wi-Fi based data transfer, PID based fan and pump control, and collecting data from sensors.
- Software modules were written for dashboards to show real-time data and subscribe to MQTT topics.

I gained a lot of valuable insight into the integration of hardware and software to produce a working system during this time.

3.3.3 Testing Phase

The most involved phase of testing was where every configuration was tested in practice.

Unit Testing:

- A multimeter, oscilloscope and power supply units were then used to test each subsystem -- the sensors, relays, and the Wi-Fi units individually.

System Testing:

- Every subsystem was tested separately before being combined into the complete system testing both hardware and software:
- I tested the control logic and helped undertake tests on the water pump itself to activate the water pump below water level.
- Each fan was tested using the Pulse Width Modulation (PWM) signals. I was able to view the results on the oscilloscope - checked the PWM circuit output from the circuit board confirming it was operational.

Debugging and Simulations:

- Prior to deployment the system logic was checked using a simulation software program.
- Also, using the serial monitor was helpful for debugging mainly with real time sensor data and with bad calibration to point to or locate the issue.

3.3.4 Validation Phase

The final phase was to validate the prototypes in a real-world operating environment.

- ✓ Hydroponic Grow Box: Evaluated in a controlled environment and ensured that plants automatically received enough light and water. Then I ran long-term tests to verify that the system could work reliably without human intervention.
- ✓ IoT dashboards: Confirmed that they would work by attaching them to machines on the manufacturing floor and ensuring that what the data indicated matched the output of the machines.
- ✓ Printed Circuit Board testing: In this process the PCB was incorporated into larger systems, and then continuity tests, short circuit evidence tests, and tested all functions.

The phase of validation taught me that real-world testing uncovers issues such as overheating, sensor drift, or unplanned Wi-Fi disconnections that were not issues experienced during testing in a simulation environment.

Acquired Proficiency

By taking part in these procedures, I was able to:

- ✓ the capacity to translate engineering designs into system needs.
- ✓ Hands-on experience with hardware assembly, soldering, and PCB design.
- ✓ Abilities in IoT programming and control logic for software development.
- ✓ Assurance while utilizing expert tools for testing and debugging.

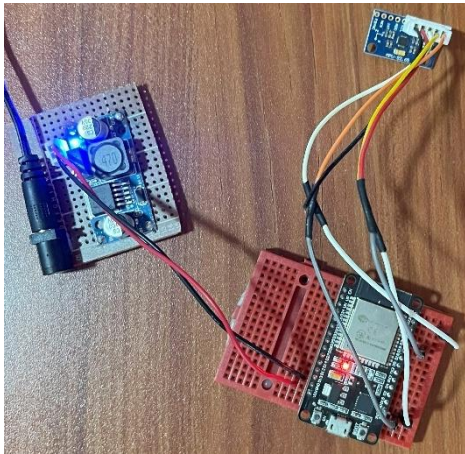


Figure 3 IoT Device ESP32 with MPU9250

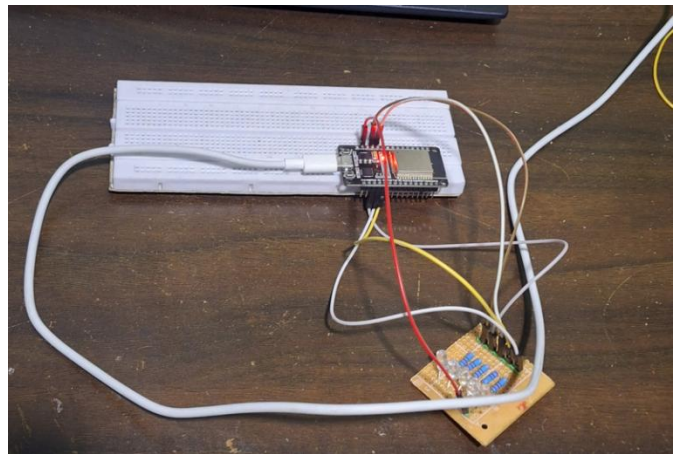


Figure 2 MQTT protocol to LED Wireless Switching Control



Figure 5 Existing Exhaust System for Hydroponic Project

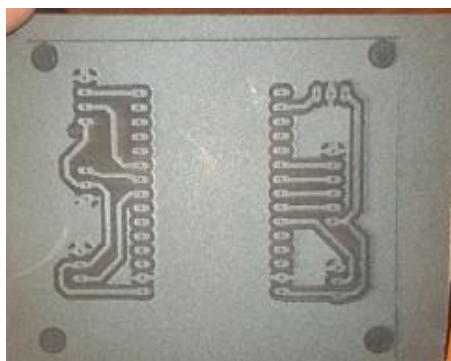


Figure 6 Single Layer Laser cut PCB for Hydroponics

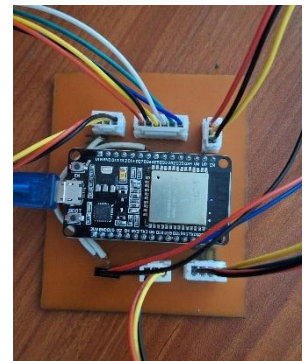


Figure 4 Soldered PCB



Figure 7 PCB Integrated System



Figure 8 Practicing Soldering Through Hole Components



Figure 10 Hot Plate setup for SMD Soldering

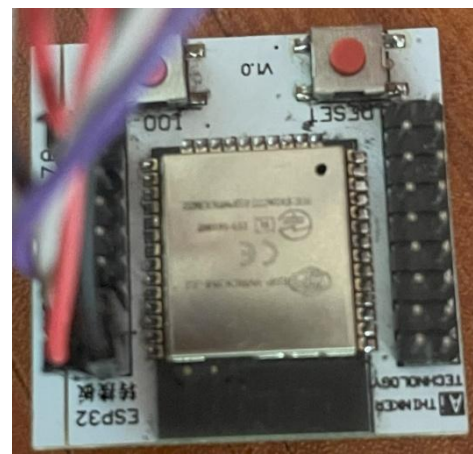


Figure 9 SMD soldered ESP32 Adapter with Module



Figure 12 Laser Cutter



Figure 11 Pressure gauge, pH and Flowmeter Replaced

3.4 Maintenance, Supervision of technical activities and Quality control.

Besides being involved in the design and development, I also observed and participated in maintenance, supervision, and in the quality assurance (QA) processes that provided dependability for industrial projects when I was training at Divor Automations Pvt. Ltd. These processes are critical for bridging the gap between creative prototypes and dependable, permanent solutions used in industries.

3.4.1 Maintenance of Systems and Hardware

Preventive maintenance ensured that all hydroponic systems, IoT dashboards, and automation hardware systems would run smoothly.

- ✓ Hydroponic Grow Box: Sensors (water level gauges and DHT22 (humidity/temperature) sensors) required calibrating periodically to maintain long-term accuracy. Occasionally, maintenance was performed to the pumps and fans to mitigate the risk of failure during normal operation.
- ✓ PCBs and Control: Assembled PCBs were regularly visual inspected for failure such as overheating, dust, and poor solder joints; in some cases, coatings and antiseptic cleaning (on higher voltage PCBs) were performed to prevent moisture damage.
- ✓ Automation Prototypes (E-Kanban, DAV 1.0): Motors and actuators were regularly lubricated and tested. Additionally, the regular maintenance for this hardware also included firmware updates that could improve performance of the system.

I learned the value of preventive maintenance in reducing equipment downtime and increasing hardware and software system life.

3.4.2 Engineering Activity Supervision

I was supervised by senior engineers, along with other junior trainees, making sure we intentionally applied technical best practices to deliver on our engineering tasks.

- ✓ PCB Design Supervision: Before being manufactured, the supervisors reviewed my EasyEDA designs to ensure the ground plane layout, track width, and clearance standards were all correct.
- ✓ Assembly Supervision: I was supervised whilst soldering the PCBs to ensure the components were properly polarized, that the SMD solder was heated correctly, and that there were no cold solder joints.
- ✓ Project Progress Monitoring: Supervisors held daily standing meetings to discuss our progress, identify delays, and suggest corrective actions.

I learned how disciplinary matters, safety, and technical standards are administered and respected in engineering.

3.4.3 Quality Control (QC) Practices

Quality control (QC) practices were integrated into each stage of the development process. The organization followed formal QC processes and informal peer reviews of several prototypes for reliability purposes.

PCB Testing:

- ✓ After soldering, continuity tests were conducted to verify proper connections.
- ✓ Short-circuit checks verified that the device would not short-circuit upon installation and then subsequently applied power.
- ✓ Test rigs verified whether the PCB operated as requested under load conditions.

Hydroponics Grow Box QC:

- ✓ Sensor output was compared to calibrate against reference instruments.
- ✓ The system was monitored and tested over an extended period of time for reliability in changing conditions.
- ✓ Logs were kept on sensor readings and deviations over time.

Industrial Automation QC:

- ✓ DAV 1.0 was then field-tested into the application, while the safety sensors were validated against other industry standards before being deployed internationally.
- ✓ The E-Kanban systems were validated against RFID (tag and reader) detection reliability and data transmission between nodes.

Implementing these QC checks into my design process demonstrated the importance of validating your work systematically and documenting it to avoid failures in an industrial setting.

3.4.4 Reporting and Documentation

I observed a complete procedure for the generation of test reports and maintenance records, which is all part of quality control. Among the documents produced are:

- ✓ Revisions and versions of the schematic (from EasyEDA).
- ✓ Test results for each module (sensors, actuators, communication).
- ✓ With failures, it is appropriate to take corrective action.

This illustrates how documentation serves both as a technical reference, and in documenting quality control efforts.

Hydroponics Project:

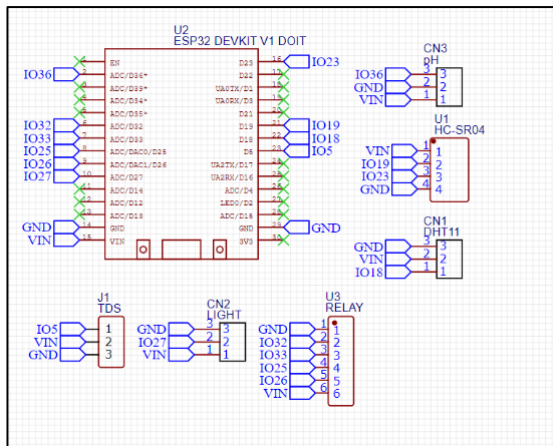


Figure 14 Schematic of Hydroponics System

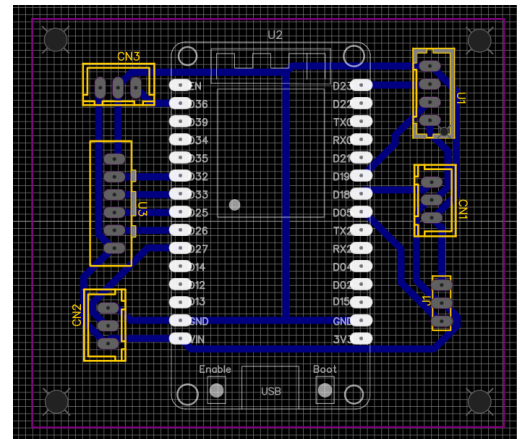


Figure 13 PCB Layout of Schematic

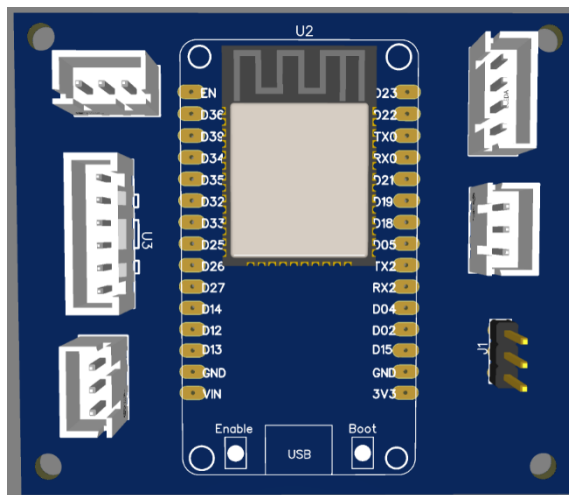


Figure 15 3D Layout of Schematic

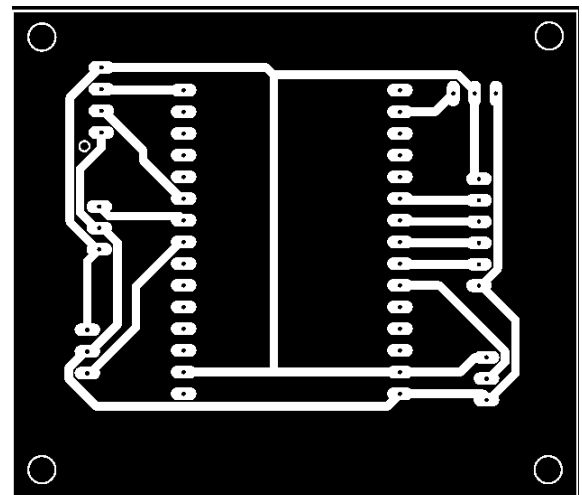


Figure 16 Mirror PNG of Traces for Laser Cutting

Timer Circuit:

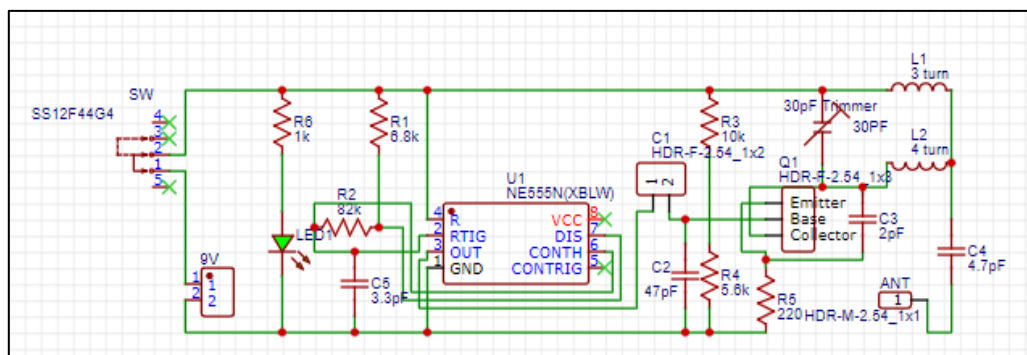


Figure 17 Timer Circuit Schematic New Project

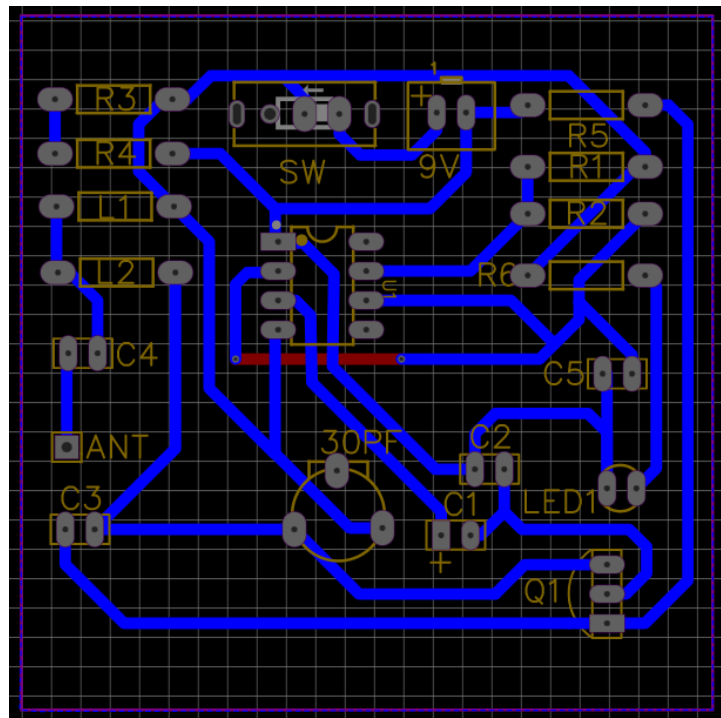


Figure 18 PCB Layout of Timer

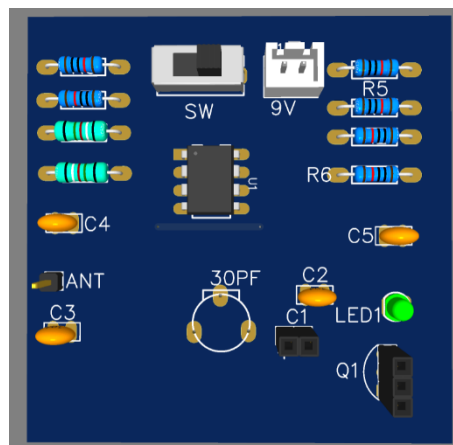


Figure 19 3D Layout Timer

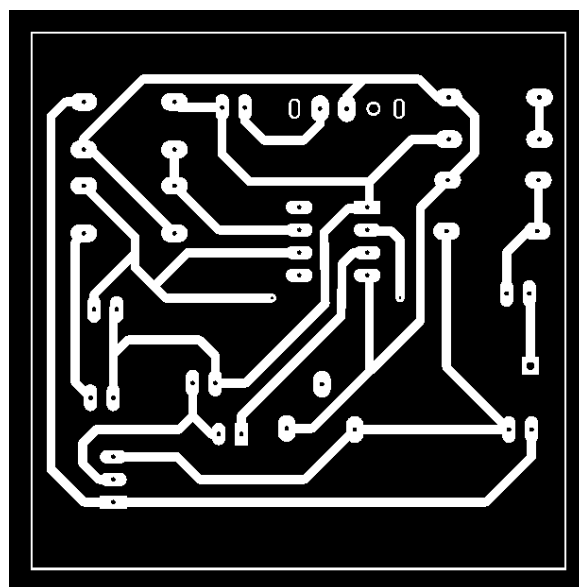


Figure 20 Mirror Trace of Timer for Laser Cutting

3.5 Hands-on experience gained through hardware/software tools, test equipment, test instruments, etc.

One of the most helpful aspects of my industrial training at Divor Automations Pvt. Ltd. was using a wide range of hardware solutions and software programs, as well as test and measurement equipment. In addition to enhancing my confidence in a technical setting through hands-on experience, this helped me develop the habit of selecting and using the most effective tool for a particular engineering problem.

3.5.1 Hardware Tools

I accumulated hands-on experience with numerous hardware platforms and electronic components:

- ✓ **ESP32 Microcontroller:** This microcontroller was used heavily for both the Hydroponics Grow Box and the IoT dashboard systems. I wrote code to gather data from sensors, send data via Wi-Fi, and control actuators. The dual-core processing capabilities of the ESP32 and built-in Wi-Fi made it a great option for real-time IoT applications.
- ✓ **Arduino Boards:** I used these boards for small automation projects and rapid prototyping, where coding ease was often a priority, and simplicity was needed for the task.
- ✓ **ARM Processors:** I gained observation experience with the use of ARM processors and motion controllers for advanced automation projects that required more engaged computational power.
- ✓ **Sensors and Actuators:** I worked with multiple sensors both DHT22 (humidity/temperature), ultrasonic sensors, pH sensors, and relay-controlled water pumps. Each requires calibration, wiring of the projects, and integration with various control boards.
- ✓ **Soldering Stations & Hot Plate for SMD:** I completed the practice of soldering both through-hole components and surface-mount components, as I learned even more about my precision and thermal management during assembly.
- ✓ **PCB Fabrication Kits:** Prototype PCBs were created using the design developed in EasyEDA (etching, drilling, and mounting the components).

All these tools have therefore allowed me to go from conceptualizing a circuit design to a working hardware prototype.

3.5.2 Software Tools

Proficiency in a range of software tools to facilitate design, simulation, and coding was taught during the training.

- ✓ EasyEDA CAD: I used EasyEDA to design PCB schematics, manage routing tracks, then generate Gerber files. I learned that appropriate placement of components, trace width, and ground plane design is important.
- ✓ Arduino IDE: Used for programming ESP32 and Arduino boards, I was able to write modular code that would take input from sensors, drive actuators, and communicate with cloud servers.
- ✓ Python: This was used to handle data and develop dashboards. I used Python for data visualization, MQTT client, and the simulation of IoT devices.
- ✓ SCADA/HMI Platforms: I learned how industrial monitoring systems are developed and customized for clients. These platforms allowed me to understand data visualization and process control in factories.

Through these software tools I was able to understand how the software integrates with hardware to create full automation solutions.

3.5.3 Test Tool

Each project contained testing as an important aspect. I utilized several types of test tools to verify that the systems were operating correctly:

- ✓ Digital oscilloscope: The digital oscilloscope enabled viewing of communication signals between modules, as well as confirmation of PWM signals generated by the ESP32 for fan speed control.
- ✓ Power Supply Units: Movable DC power supplies were also utilized to test circuits in varying voltage/current scenarios.
- ✓ Color coded multimeters are used for measuring resistance, continuity, and simple voltage/current verification during PCB assembly and troubleshooting.
- ✓ Spectrum Analyzer: The spectrum analyzer was used for confirming the ESP32 modules' wireless connection and signal strength on Internet of Things projects.

The experience also gave me a deeper understanding of the role of precision measurement instruments for locating and resolving technical issues.

3.5.4 Test Equipment and Experimental Setups

Along with standard test equipment, I used several specialized equipment and setups:

- ✓ Calibration Kits for Sensors: To assure accurate environmental monitoring by calibrating pH meters and DHT22 sensors for the hydroponics system.
- ✓ Motor test setups, during the automated vehicle DAV 1.0, were used to measure the torque, speeds, and stability of motor drivers.
- ✓ Environmental test setups, in the Grow Box, were used to emulate different humidity levels, water levels, and temperatures. This allowed the system to be evaluated for long duration stress.
- ✓ Laser Cutter - Informed knowledge of laser cutting for PCB prototypes and enclosures.

This equipment showed the importance of assured testing environments in validating reliability of systems.

3.5.5 Gained Competence and Foundational Skills

Being proficient with these tools and methods allowed me to:

- ✓ Utilize oscilloscopes, multimeters, and serial monitors to better your diagnostic skills.
- ✓ Use soldering stations and hot plates to improve the accuracy of through-hole and SMD PCB soldering.
- ✓ Design with practicality by understanding how your PCB design choices in EasyEDA will affect the real-world function of the hardware.
- ✓ Improve your programming skills, especially concerning embedded systems (ESP32 and Arduino).
- ✓ Develop a process of troubleshooting that allowed me to use measurement tools in a systematic manner to discover and resolve the issue.

Overall, this hands-on experience gave me knowledge on how to prepare for future roles in industrial automation and embedded systems by bringing together theory and practical engineering skills.

3.6 Problems encountered and how they were resolved.

Whether the engineering project is related to class or practice, it will face real-world problems outside of what was considered during the design process. At Divor Automations Pvt. Ltd., I faced numerous issues during my training with software integration, hardware design, and testing, all of which were valuable opportunities to develop my skills in critical thinking and systematic troubleshooting.

3.6.1 Sensor Calibration Issues

Issue: Temperature and humidity sensors, such as DHT22, frequently provided inconsistent readings in the Hydroponics Grow Box. Even with everything held constant, humidity records fluctuated by $\pm 5\%$, resulting in load actions (for example, causing fans to turn on unnecessarily).

Root cause:

- ✓ Electrical noise in the environment interfered with sensor signals.
- ✓ The original code did not use adequate software filtering.

Solution:

- ✓ We added moving average filters to the ESP32 code to smooth out sensor data.
- ✓ Re-set sensors in a controlled lab environment using reference equipment.
- ✓ Sensor wiring was completed with shielded cables to decrease electrical interference.

Lesson Learned:

- ✓ System reliability is dependent on properly calibrated sensors. To ensure clean data, good software filtering and hardware shielding are necessary.

3.6.2 Inconsistent Power Supply

Problem: During PCB testing, certain modules occasionally restarted when all actuators (pump + fan) were powered on at the same time.

Root Cause:

- ✓ The linear regulator used in the original power supply design was unable to accommodate sudden bursts of current.
- ✓ The use of decoupling capacitors was insufficient.

Solution:

- ✓ The power circuit was redesigned using a buck converter to provide a stable 5V from the 12V power supply.
- ✓ 100 μ F electrolytic capacitors and 0.1 μ F ceramic capacitors were added close to the ICs to resist transients and filter noise.
- ✓ The system was tested under load before release.

Lessons Learned:

- ✓ Power supply design is the foundation of any electronic system. The automation and Internet of Things worlds require switching regulators and decoupling capacitors to achieve stability.

3.6.3 MQTT & Wi-Fi Communication Problems 3.1 MQTT & Wi-Fi Communication Problem

In the implementation of the IoT Dashboard and Hydroponics Grow Box projects, the ESP32 modules sometimes lost Wi-Fi connection which resulted in timed loss of the ability for real-time monitoring via the dashboard.

Root Cause:

Some areas of the laboratory have poor Wi-Fi signal strength.

Automatics reconnect was not included in the original firmware.

Solution:

To enable an automatic restart of Wi-Fi connection should it drop, we added a watchdog timer to the ESP32 firmware.

We added auto-reconnect logic to the MQTT client code.

Wi-Fi range extenders were used to ensure signal coverage in all testing areas.

Lesson Learned:

All IoT systems should be designed with redundancy and self-healing mechanisms in order to recover from communications failures.

3.6.4 The Hydroponics Control Box Gets Too Hot

An issue arose with the casing of the hydroponics controller (ESP32, relays, buck converter and sensors), that had gotten too hot after prolonged use. The interior of the box reached temperatures above 45°C, which increases the chance of failure of the components.

Root cause:

- ✓ Inadequate ventilation in the enclosure.
- ✓ Relays and power components generate heat.

Solution:

- ✓ Used tiny 5 V cooling fans to aggressively dissipate heat.
- ✓ Added ventilation slits to the enclosure design with a laser cutter.
- ✓ In the software, optimized to prevent the relays from being actuated unnecessarily, reducing heat build-up.

Lessons Learned:

- ✓ A major element of embedded system design is thermal management. Reliability testing in the field will reveal latent heat problems that may not show up in a shorter test period where the circuits appear to be working fine.

3.6.5 PCB Assembly Issues

Findings: A few early PCB prototypes had short circuits and ultimately failed after power-up.

Root Cause:

- ✓ The components were hand-soldered in the incorrect locations.
- ✓ Hot-plate reflow soldering bridges small-pitch SMD components.

Solution:

- ✓ EasyEDA was utilized to check the layout beforehand using a design rule check (DRC) to help minimize layout errors.
- ✓ Multimeter continuity tests were introduced after every step of construction.
- ✓ Soldering skills were practiced under supervision and gradually improved through focused practice.

Lessons learned:

- ✓ Even a small assembly mistake can damage an entire PCB. A lot of time and resources can be saved by carrying out thorough testing at every step (design → layout → solder).

4.0 ACQUIRED SOFT SKILLS

Divor Automations Pvt. Ltd. provided an industrial training experience that not only focused on technical skills and project opportunities but also was vital to enhancing my soft skills, which are foundational for effective collaboration and teamwork, leadership, and for the success of my professional development in the long term. The training environment helped foster interaction, team dynamics, and ongoing development of each participant's management and communication skills.

4.1 Soft Skills Gained

During my training, I gained several important soft skills that will contribute to my personal and professional development:

Communication Skills

- ✓ Enhanced my ability to communicate around technical concepts with engineers and non-technical personnel.
- ✓ Gained confidence communicating project updates at weekly review meetings.
- ✓ Understood the value of writing clear documentation and reports to pass on institutional knowledge

Teamwork and Collaboration

- ✓ Worked on multi-disciplinary teams of engineers, designers and technical staff/technicians.
- ✓ Learned how to share responsibilities effectively, work through disagreements and collaborate to meet project deliverables.
- ✓ Understood the importance of engaging with multiple perspectives when attempting to solve problems.

Time Management

- ✓ Learned how to prioritize multiple projects, particularly when managing work on more than one project (Hydroponics, IoT dashboards, PCB design).
- ✓ I learned to navigate deadlines on projects and observed where my supervisory personnel were placed tasks and assignments in a way that created efficiencies.

Problem Solving Mindset

- ✓ Developed skills to think through problems with a scaffolded mindset as opposed to thinking in a reactive manner.
- ✓ Gained analytic thought processes and troubleshooting abilities in a real-world application context with projects such as sensor calibration, issues with PCB faults.

Adaptability and Flexibility

- ✓ Adapted very quickly when transitioning to using new tools and technologies over short periods of time, including but not limited to the EasyEDA PCB design software and MQTT protocols for communication.
- ✓ Learned to operate under some level of duress when issues were experienced during a project phase.

4.2 Soft Skill Training programs underwent.

In addition to my training, I was able to attend scheduled soft skill development experiences facilitated by the company during the process of training:

Technical Presentation Workshops

- ✓ During the program, I was instructed in sessions in which trainees needed to prepare PowerPoint presentations for their assigned projects.
- ✓ This enhanced my ability to communicate technical aspects coherently and effectively to present in a professional manner.

Leadership & Team-Building Activities

- ✓ Some group activities, including project brainstorming sessions, were conducted to develop leadership style of work and group cohesion in team objectives.
- ✓ I learned to be proactive in small group objectives, which enhanced my ability to make sound decisions.

Professional Etiquette Training

- ✓ Sessions were facilitated to stress the status of professional behavior, email etiquette, and workplace interaction.
- ✓ This helped me understand the expectations of behavior in corporate environments.

Routine Project Progress Reviews

- ✓ The weekly review meetings also served as an informal but informative training program where I learned how to accept constructive feedback.
- ✓ Also, I learned to enhance my listening skills since it was important to successfully implement suggestions from senior engineers for ensuring the projects continued forward.

4.3 Soft skill development activities

In addition to formal training sessions, several of the day-to-day activities that I experienced during the training were also effective avenues for developing softer skills:

Daily Morning Standup Meetings

- ✓ Acting in the capacity of a participant during morning meetings allowed me to grow my confidence in "to-the-point" communication.
- ✓ I had the opportunity to not only communicate progress, but to raise potential red flag issues, and ask for assistance when necessary.

Documentation and Report Writing

- ✓ Completion of project documentation improved my technical writing skills by ensuring that any documentation or reporting was clear and right to the point.
- ✓ Again, I participated in the writing of small user guides for IoT devices, which required writing in plain language suitable for the clients.

Peer to Peer Knowledge Sharing

- ✓ I participated in informal discussions with fellow trainees and senior engineers.
- ✓ This improved my listening, questioning, and mentoring skills.

Client Interaction (Observations)

- ✓ While I didn't interact co-directly with the client, I was an observer when the project manager and senior engineer did in fact meet with the client.
- ✓ I learned the values around professionalism, patience, and active listening when interacting in a client environment.

Problem-Solving Sessions

- ✓ Whenever there were technical issues (i.e., power instability, Wi-Fi failures), a brainstorming session was held.
- ✓ This is where my analytical skills grew and my confidence improved by being able to contribute a red flag, issue, or solutions in a group session.

5.0 PRACTICE OF PROFESSIONAL STANDARDS AND ENGINEERING ETHICS, AND ENVIRONMENT & SUSTAINABILITY

Throughout my training with Divor Automations Pvt. Ltd., I observed and took part in several activities that addressed professionalism, engineering ethics, safety practices, sustainability ideas, and technological excellence. In addition to solving technical problems, engineers must make ethically sound, environmentally conscious, and sustainable decisions for future generations.

5.1 Orientation Program, additional studies and other related training programs.

Before starting on any project work, all trainees, me included, took part in an orientation course. Through the course, we became familiar with:

- ✓ The organization's expectations, workplace culture, and code of conduct.
- ✓ Fundamentals of intellectual property protection, emphasizing that all work was proprietary and confidential.
- ✓ Lab health and safety procedures such as equipment handling procedures and fire safety drills.

During my orientation, I was further encouraged to engage with additional workshops and self-directed work to improve my understanding of:

- ✓ Internet of Things communications protocols (MQTT, Wi-Fi, RFID).
- ✓ EasyEDA PCB design, which would be vital to my involvement in hardware projects.
- ✓ Environmental management of hydroponics systems, which has developed my understanding of sustainable farming practices.

These studies provided me with the technical expertise and ethical awareness required in a professional engineering environment.

5.2 Ethical considerations in decision-making processes.

When making engineering decisions, ethical considerations are often an aspect, particularly when it relates to safety, cost or sustainability. Some of the ethical considerations I observed and practiced are:

- ✓ Client Centered Design: Rather than cut corners to meet deadlines for the design of IoT dashboards and automation systems, the engineers implemented the client's wishes in their designs.
- ✓ Data Integrity: Stressing the accuracy of sensor reporting results. I learned that the falsification or exaggeration of data is unethical and harms both the client as well as society.

- ✓ Respect to Intellectual Property: There was an understanding that designs and codes were treated as property of the company and as rehired or contracted individuals we were reminded not to disclose or share them.
- ✓ Use of Resources: Materials or components provided for a project were to be used responsibly and not wasted unless necessary.

I realized from my experiences with the aforementioned situations, that ethics in engineering is not an afterthought, it is a part of an engineer's duty that holds the trust between an engineer, the client and society.

5.3 Health, Safety, and Environment (HSE) regulations and best practices in engineering.

Throughout my research placement, I was frequently reminded of the importance of Health & Safety (HSE):

- ✓ Electrical Safety: We always adhered to lock-out/tag-out procedures when testing high-voltage circuits. We were required to use insulation gloves and multimeters rated for the voltages we were working with to minimize the risk of injury.
- ✓ Soldering Safety: During our work with soldering irons and hot plates during SMD assembly (surface mount device assembly), we were always required to use working ventilation systems or fume extractors to minimize inhalation exposure to the soldering fumes.
- ✓ Fire Safety: All labs had fire extinguishers present, and we were advised of the emergency evacuation protocols during orientation.
- ✓ Chemical Handling: When we carried out PCB preparation using cleaning solvents, we were required to wear gloves and masks.
- ✓ Environmental safety: E-waste - defective PCBs, used components - were separated out to ensure proper disposal versus being put in regular waste.

These enforced protocols allowed me to learn that engineering safety protocols protect not just people but protect the planet as well.

5.4 Applicable Codes, Standards, and References in Engineering Applications.

The engineering profession requires conformity to international and local standards in order to ensure safety, interoperability, and quality. Some codes and standards I encountered include the following:

- ✓ ISO 9001 (Quality Management Systems) ensures that any projects are properly documented in a compliant way, tested adequately, and checked for quality overall.
- ✓ IPC Standards for PCB Design and Assembly clarify track width, soldering quality, and common processes for components placed in PCB fabrication.
- ✓ IEEE Standards for IoT and Communication Protocols generally form a best practice for wireless communication, data integrity of devices, and interoperability of devices.
- ✓ OSH standards, Occupational Safety and Health, apply to laboratory safety and worker safety overall.
- ✓ For Local Sri Lankan Standards, the company project projects that I carried out complied with the Sri Lanka Standards Institution (SLSI) for industrial automation equipment.

Overall, these standards provided an opportunity for me to comprehend and understand how engineering work must cater to globally accepted practice.

5.5 Sustainable development principles and goals.

Sustainability was a frequent topic of conversation throughout the projects and in the company's value statements. In particular, the UN Sustainable Development Goals (SDGs) below seemed especially relevant:

- ✓ **SDG 7: Affordable and Clean Energy** - The Hydroponics systems offer a solution that optimizes and reduces energy consumption.
- ✓ **SDG 9: Infrastructure, Industry, and Innovation** - By promoting autonomous solutions backed by IoT, I felt it would modernize the local industry.
- ✓ **SDG 12: Responsible Consumption and Production** - We provide our clients, through the E-Kanban system, with assurances that we utilize resources to the best of our ability, and we significantly reduce the use of paper.
- ✓ **SDG 13: Climate Action** - The use of automated technologies to optimize energy consumption would reduce our environmental impact.

It became apparent to me that every design choice an engineer makes can either have a positive or negative impact on sustainability, and that sustainability is a core responsibility of an engineering professional.

5.6 Sustainability in engineering projects (energy efficiency, waste reduction, eco-friendly materials) including Environmental impact assessments and sustainable design concepts.

The projects I developed demonstrated a strong focus on eco-friendly and sustainable design principles:

Energy Efficiency:

- ✓ Hydroponics Grow Box was engineered to work with low-power IoT sensors and optimized lighting systems (LED grow lights) to conserve energy.
- ✓ The IoT dashboards were cloud-based solutions that did not rely on large, physical servers.

Waste Reduction:

- ✓ The E-Kanban systems minimized the use of paper in the factory and reduced waste while increasing resource efficiency.
- ✓ In PCB assembly, the engineer accounted for all components and eliminated waste.

Eco-Friendly Materials:

- ✓ The PCB prototyping used lead-free solder that met RoHS standards.
- ✓ The enclosures for the devices were made with durable and recyclable materials.

Environmental Impact Assessments (EIA):

- ✓ Before the engineers implemented the automation systems, they evaluated energy usage, waste management, and potential hazards.
- ✓ In hydroponics, they focused on re-use of water and limited use of chemicals, resulting in less environmental damage as compared to traditional farming methods.

All these examples illustrate how engineering and sustainability are connected and that an engineer can consider environmental impacts at each phase of the process. Divor Automated ensure their products supported the long-term preservation of nature.

6.0 SUMMARY AND CONCLUSIONS

The industrial training at Divor Automations Pvt. Ltd. was an extremely valuable experience that enabled me to connect the skills and knowledge obtained at university with actual practice in the industry. During my training, I was fortunate to work on many projects including IoT systems, PCB design, and prototypes for industrial automation. I learned about many different topics and therefore the combination of all these experiences provided me with a well-rounded perspective on how theoretical engineering concepts can be implemented into solutions that address real-world problems.

One of the most valuable projects I was involved in was the Hydroponics Grow Box, which integrates IoT applications into sustainable agriculture. This project provided me with an application of electronics, control systems, and embedded programming knowledge that led to the development of an automated system for growing plants. Similarly, my work on IA dashboards and E-Kanban systems provided valuable insight on how data acquisition, monitoring and visualization can improve industrial efficiency. Not only did I gain further technical experience and skill development, but the project assignments also exposed me to problem solving under many different and dynamic environments.

The training also emphasized the importance of professionalism, engineering ethics, and sustainability. I saw firsthand how engineering decisions were made based on ethical considerations, a client's request, and international codes and standards. Safety procedures were considered priority, followed closely by quality control and environmental sustainability, while ensuring engineering solutions were dependable, safe, and environmentally friendly. This helped me (further) recognize the duty engineers have to society for the innovative solutions they provide but also to allow for solutions to maintain ethics and sustainability.

The training demonstrated soft skills beyond technical knowledge, such as communication, collaboration, adaptability and time management. I gained these skills in daily interactions with supervisors, during team-based projects, and in presentations during team review meetings. Participating in these activities fostered my confidence to collaborate in a professional manner and to bring structure, discipline, and professionalism to the responsibilities I was given.

In summary, this industrial training has been a transformative process that improved my technical knowledge, my professional and interpersonal skills and taught me the importance of ethics and sustainability in engineering. The exposure I received at Divor Automations has prepared me for future engineering challenges with competency, responsibility, and confidence. The experience I gained in this period will provide me with a strong foundation as I begin my career as an engineer and propel me to contribute both to industry and to society at large.

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